Electricity Demand Forecasting Using AI

Project Lab Automation

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Motivation

Problems:

- Better forecasts reduce imbalance penalties and operational inefficiencies [1]
- Increasing variability from renewables makes accurate planning more challenging

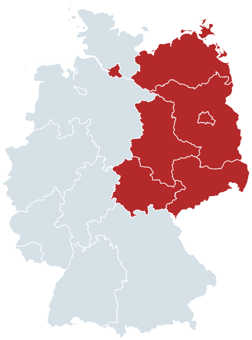


Figure 1: 50Hertz Region.

Motivation

Objectives:

- Build a complete dataset
- Forecast the hourly demand profile for the following day

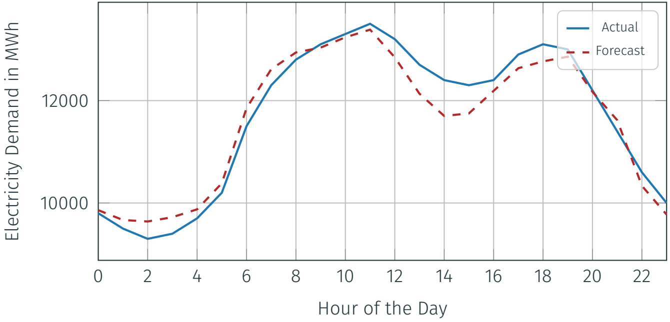


Figure 2: Example of hourly electricity demand and forecast.

Dataset

Time range: 01/01/2015–12/31/2024 with hourly resolution
Power Sources: From the Federal Network Agency, freely available (CC BY 4.0) [2]
Weather: Five different stations across region using Meteostat [3]
Temporal Features: School holidays across all federal states [4]
Market Data: Day-ahead Price in DE/(AT)/LU bidding zone [5]

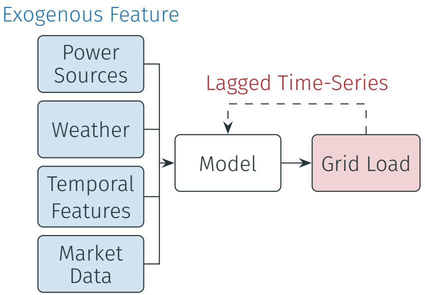


Figure 3: Forecasting model with exogenous features and lagged targets.

Dataset

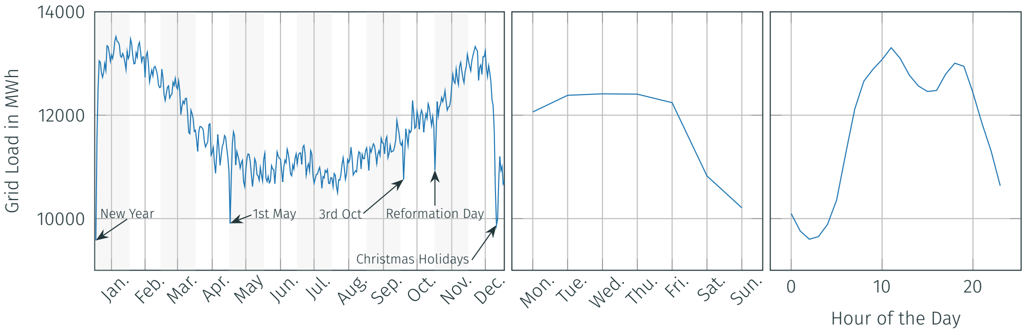


Figure 4: Temporal patterns of average grid load.

Backtesting

Backtesting setup:

- Forecast horizon: 24 hours
- Model is not refitted at each fold
- Validation period: average MAPE in 01/01/2024–12/31/2024

Metric	Unit	Description
MSE	MWh^2	Penalizes large errors more strongly for detecting instability in forecasts
MAPE	%	Relative error measure comparable across different demand levels [6]

Baseline Model

Seasonal Naive Forecaster uses equivalent point in the past “season”, e.g. the same hour on the previous day [7]

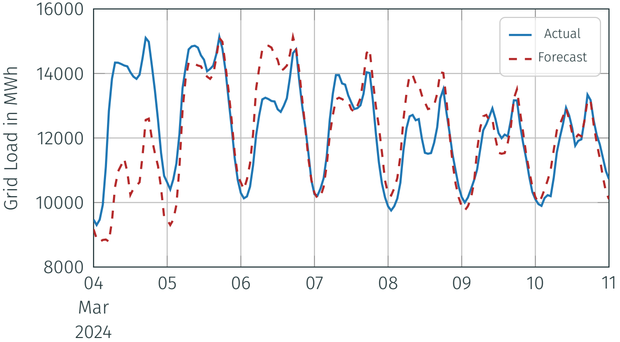
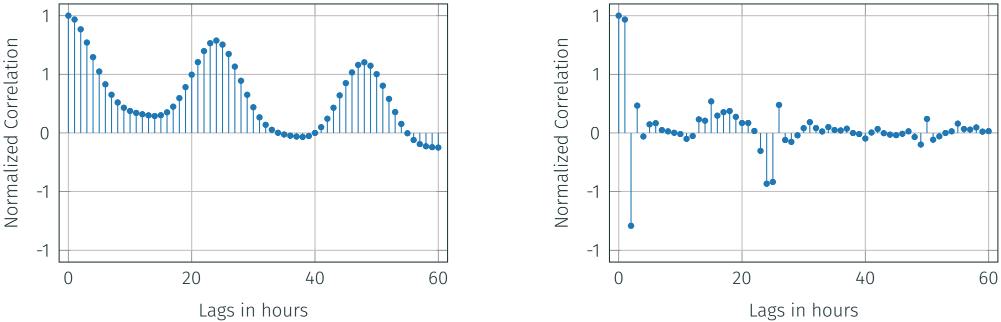


Figure 5: Baseline model with MAPE of 8.57%.



(a) Autocorrelation. **(b)** Partial Autocorrelation.

Figure 6: Autocorrelation and partial autocorrelation of the grid load time series.

Forecasting Model – Recursive Forecaster

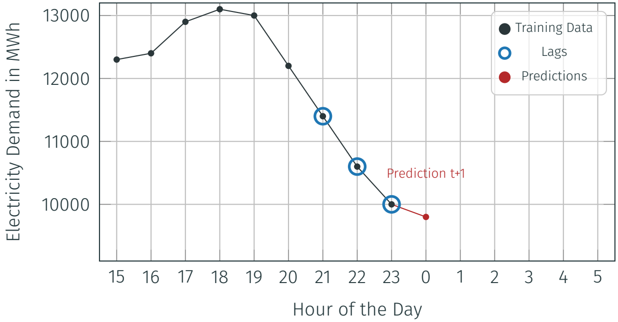
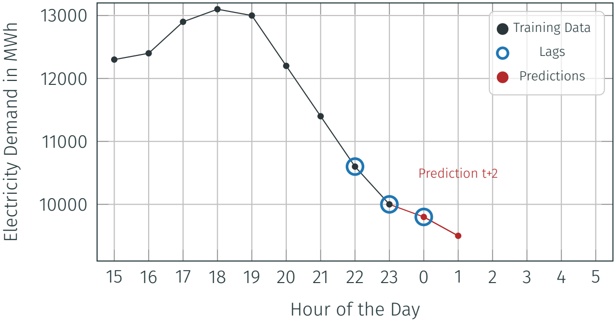


Figure 7: Recursive multi-step forecasting.

Forecasting Model – Recursive Forecaster



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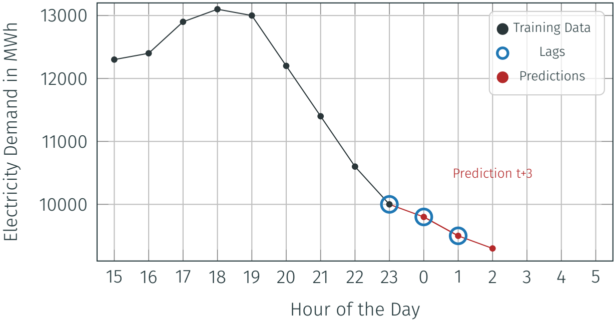

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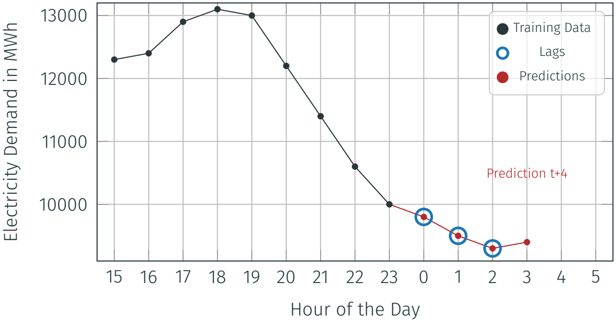

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Forecasting Model – Recursive Forecaster

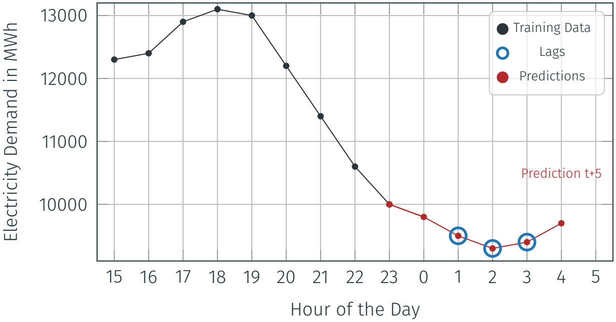


Figure 7: Recursive multi-step forecasting.

Estimator

LightGBM

- Histogram-based splitting for high efficiency
- Leaf-wise tree growth strategy [8]

XGBoost

- Strong regularization to reduce overfitting
- Level-wise tree growth for stable training [9]

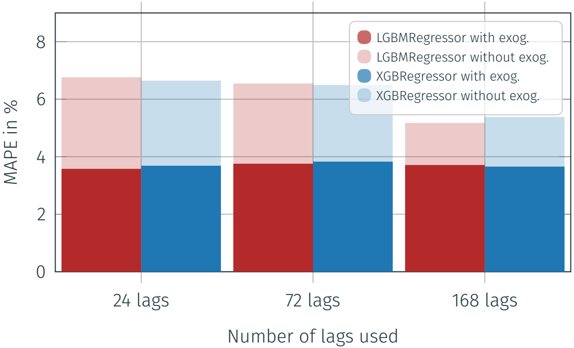


Figure 8: Estimator comparison with exogenous features and different lags.

Hyperparameter Tuning

Computational Cost:

- XGBoost: ∼60 min
- LightGBM: ∼58 min

Performance Gain:

- XGBoost: MAPE from 3.66% to 3.15%
- LightGBM: MAPE from 3.71% to 3.16%

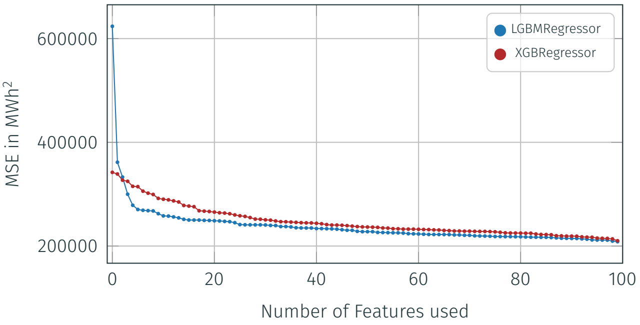


Figure 9: Bayesian Optimization with 100 trials to efficiently explore the hyperparameter space.

Final Model

Model is refitted at each iteration to reflect real-world production behavior

Backtesting time with refitting increases up to 41min (with 2 CPU Cores)

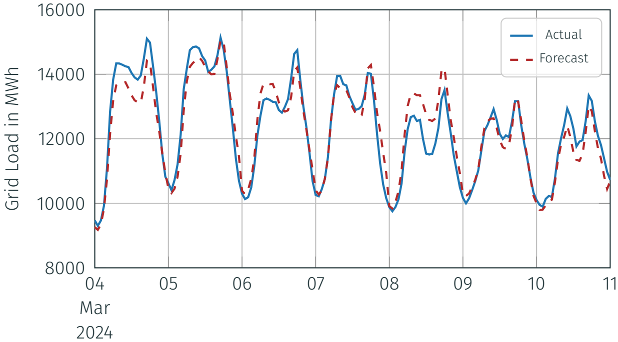
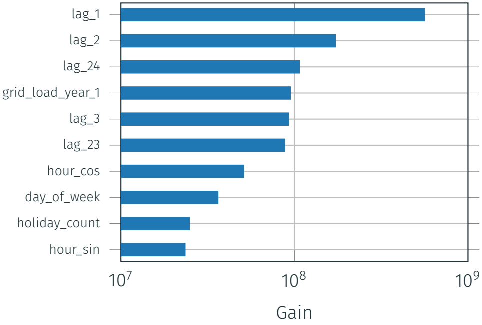


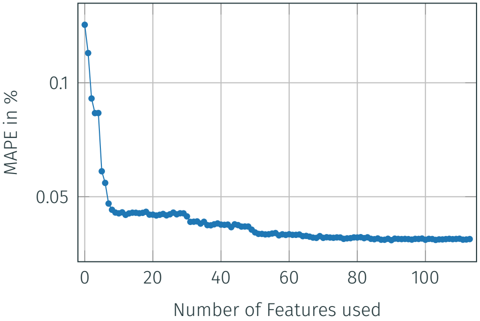
Figure 10: Tuned XGBRegressor with a MAPE of 2.94%.

Feature Importance

Gain importance: Total reduction of the loss function achieved by splits using the feature



(a) Feature importance of forecasting model.



(b) MAPE as a function of the number of selected features using rank-based feature addition.

Figure 11: Feature Importance of XGBRegressor.

Outlook

- Explore other features that might improved model performance, e.g. polynomial features
- Explore probabilistic forecasting approaches to better quantify uncertainty
- Investigate deep learning–based models for capturing more complex temporal patterns

Questions or Remarks?

[1] A. Miraki, P. Parviainen, and R. Arghandeh, *Electricity demand forecasting at distribution and household levels using explainable causal graph neural network*, *Energy and AI*, vol. 16, p. 100 368, Apr. 2024. DOI: *https://doi.org/10.1016/j.egyai.2024.100368*.

[2] Bundesnetzagentur, *SMARD – Strommarktdaten für Deutschland*, smard.de, Accessed: Nov. 13, 2025, 2025.

[3] C. S. Lamprecht, *Meteostat Python*, meteostat.net, Accessed: November 12, 2025, 2020.

[4] M. N. Johannsen and C. Sanders, *Schulferien und Feiertage in Deutschland*, schulferien.org, Accessed: Decemeber 01, 2025, 2001.

[5] Burger, Bruno, *Energy-Charts API Documentation*, Accessed: 2026-01-16.

[6] M. Fotakis, *The Validity of Backtesting for Evaluation of Autoregressive Time Series Predictions*, M.S. thesis, Tilburg University, 2021.

[7] J. Amat Rodrigo and J. Escobar Ortiz, *skforecast*, version 0.19.1, Dec. 2025. DOI: *10.5281/zenodo.8382788*.

[8] Y. Shi, G. Ke, D. Soukhavong, J. Lamb, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, et al., lightgbm: Light Gradient Boosting Machine, 2025.

[9] T. Chen and C. Guestrin, *XGBoost: A Scalable Tree Boosting System*, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, New York, NY, USA: Association for Computing Machinery, 2016, pp. 785–794. DOI: *10.1145/2939672.2939785*.

[10] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. New York, NY, USA: Springer New York Inc., 2001. DOI: *10.1007/b94608*.

[11] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., *Scikit-learn: Machine learning in Python*, *Journal of Machine Learning Research*, vol. 12, no. Oct, pp. 2825–2830, 2011. DOI: *10.5555/1953048.2078195*.

[12] R. L. Neymeyer, R. Rennert, and A. Dilg, *Electricity Demand Forecasting using AI*, github.com/lar-tech/electricity-demand-forecasting, 2026.

Appendix – Gradient Boosting Machine

Fitting Algorithm [10]:

1. Start with simple model, e.g. mean of target variable
   $F_1(x) = f_1(x) = \bar{y}$
2. Calculate error gradient
   $\hat{r}_{mi} = -\frac{\partial \mathcal{L}(y_i, F(x_i))}{\partial F(x_i)}\Big|_{F=F_m} \quad \forall i \in \{1, \dots, N\}]$
3. Fit a new weak learner to the residuals
   $f_{m+1}(x) \approx \hat{r}_m$
4. Determine how much of the new learner to add
   $\gamma_{m+1} = \arg\min_\gamma \sum_{i=1}^{N} \mathcal{L}(y_i, F_m(x_i) + \gamma f_{m+1}(x_i))$
5. Update the model
   $F_{m+1}(x) = F_m(x) + \gamma_{m+1} f_{m+1}(x)$

LightGBM uses leaf-wise tree growth, leaf with the highest loss reduction is split [8]

Appendix – Cyclic Encoding

- Cyclic encoding represents time features using sine and cosine functions [11]
- Preserves continuity, e.g. the distance between 23:00 and 00:00 equals that between consecutive hours
- Helps the model capture periodic temporal patterns

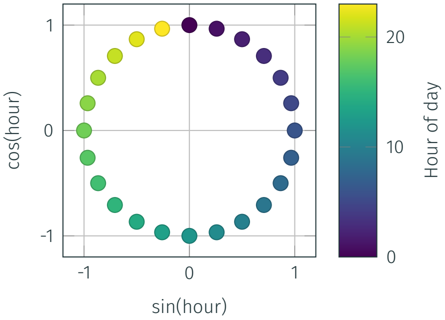


Figure 12: Cyclic Encoding of Hour Feature

Appendix – SHapley Additive exPlanations Values

- SHAP values quantify how much each feature contributes to an individual prediction compared to a baseline.
- They are based on concepts from cooperative game theory and provide consistent and interpretable feature attributions.

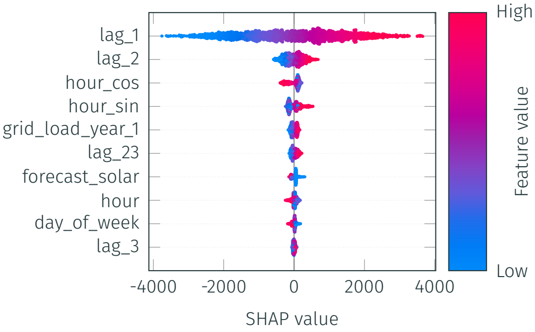


Figure 13: SHAP summary of the most important features on the model output.